

The segmentation table contains selected retained conditions rather than a complete corruption benchmark. It cannot establish improvement for unreported conditions, and its values are not directly comparable with classification corruption results.

Appendix I - Mobile Interface and Deployment Configuration

This appendix records the final conditional two-model Android deployment contract and the observational runtime protocol. The mobile evidence is operational feasibility evidence, not field disease-accuracy validation.

Table I.1. Final mobile model-bundle contract

Role	Deployed artifact / precision	Interface contract	Verified boundary
Classifier	yolo26m_asl_ldam_simam_dcfr_fp32.tflite; FP32	Input [1,224,224,3]; output [1,4]; order Healthy, BG, WSSV, WSSV_BG	Always executes first. Export size/hash are retained by the classification weight audit, which also flags an internal train-args naming conflict; export identity alone does not prove the final reported metrics.
Segmenter	yolo11n_seg_float16.tflite; FP16; retained size 5.65 MB	YOLO11n-seg + CA→SimAM at P3/P4/P5; 640 × 640 RGB; BG/WSSV instance masks	Invoked only for above-threshold BG/WSSV/WSSV_BG outputs. The current document set retains final artifact metadata and app usage, but no segmentation cryptographic hash is retained here.
Branch control	Application-side dispatch	Healthy or below threshold → stop; disease above threshold → segmentation	Predicted class controls execution and is not a verified field disease label.

The final classifier parameter count is not copied from the plain YOLO26m candidate row. The audited candidate export is 41,443,690 bytes with SHA-256 9834ecbaeee14878da2fd53a811005cf a2ed8192b94f9d8a7f5b732b19d1689d. The archived weight audit also preserves the train-args naming conflict, so this hash identifies the audited export but does not independently reproduce the final reported metrics.

Table I.2. Android device configuration ledger

Device	OS / processor as recorded	CPU	Memory as recorded	Storage	GPU
vivo Y100	Android 15; Snapdragon 685, reported up to 2.8 GHz	8 cores	16 GB reported configuration	128 GB	Not recorded
Samsung A32	Android 13; MediaTek Helio G80, 2.00 GHz	8 cores	6 + 6 GB reported configuration	128 GB	Not recorded
POCO M7 Pro 5G	Android 16; MediaTek Dimensity 7025 Ultra	8 cores	8 + 4 GB reported configuration	256 GB	Not recorded

Memory entries are preserved as reported/configured values. Values such as 6+6 GB and 8+4 GB are not rewritten as wholly physical RAM, and missing GPU identifiers are not guessed. All three devices used the same application build, the same model bundle, and the common configured four-thread CPU path. GPU delegate execution was disabled in the retained application configuration.

Table I.3. Runtime logging and branch semantics

Runtime item	Recorded behavior	Interpretation boundary
Classifier execution	Runs for every accepted image	Every usable runtime record includes classification.
No-seg branch	Healthy above threshold or below-threshold short circuit	No disease mask is generated.
Segmentation-triggered branch	Above-threshold BG, WSSV, or WSSV_BG dispatches the second model	Branch includes additional segmentation preprocessing/decoding/application work.
Timing fields	Application-recorded elapsed time, speed, FPS, and running averages	May include preprocessing, interpreter activity, postprocessing, mask decoding, and UI/application overhead; not pure NN latency unless independently scoped.
Ground-truth fields	ground_truth_label and is_correct left empty	No mobile field accuracy, sensitivity, specificity, prevalence, or clinical-validity claim.